INDIA INNOVATES 2026

FARM2FUTURE

Rooted In Tradition Driven By Tradition

Presented By:
Adithya K
Sanjith Sharmaa R
Hariharan R S

@farm2future.in

Farm2Future GitHub

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01.
Executive Summary

Introduction

Company Name: **Farm2Future**

Founders: Adithya K, Sanjith Sharmaa R, Hariharan R S

Github: [Farm2Future GitHub]

Website: [Farm2Future Website]

Tagline: **Rooted in Tradition, Driven by Innovation**

About Us

Who We Are

Farm2Future is an agri-tech startup transforming farming through sustainable and affordable robotic rental solutions. Founded by young innovators, we provide smart, AI-powered robots like Agribot, AgriDrone, and AgriArm that enhance productivity while reducing costs and environmental impact. Our rental-based model makes advanced technology accessible to every farmer, promoting smarter, greener, and more efficient agriculture.

Farm2Future Website

Our Vision & Mission

Movement Goal

Vision

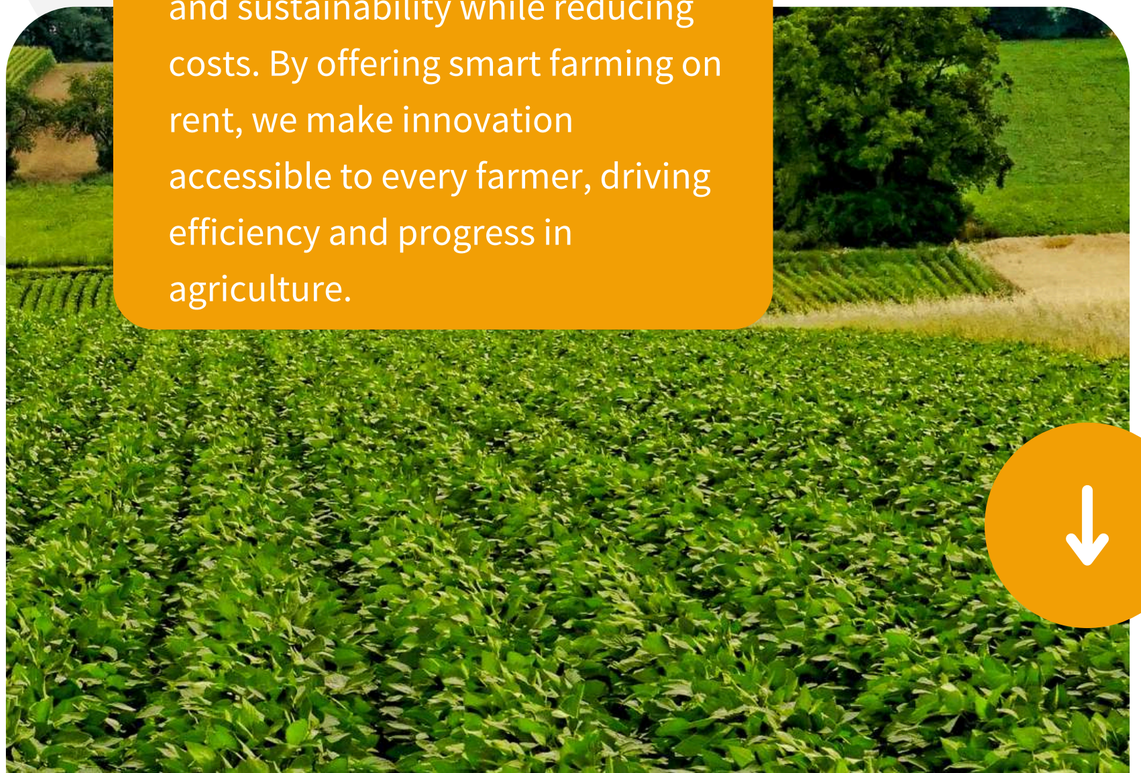
A future where technology bridges the gap for small and medium-scale farmers, driving innovation, productivity, and prosperity in rural communities.

Mission

Transforming farming with affordable, cutting-edge robotic solutions that empower farmers to achieve sustainable, efficient, and profitable agriculture.

Value

Farm2Future empowers farmers with affordable access to advanced robotics, increasing productivity and sustainability while reducing costs. By offering smart farming on rent, we make innovation accessible to every farmer, driving efficiency and progress in agriculture.



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02.
Problem Statement

Problem & Current Solutions

System-Level Agricultural Challenges

- **Rural Labor Shortages & Migration**
  Declining farm labor availability is slowing operations and increasing seasonal workforce dependency across districts.
- **Low Mechanization Penetration**
  Small and marginal farmers lack access to shared automation infrastructure, limiting productivity at scale.
- **Suboptimal Yield per Hectare**
  Traditional and labor-intensive practices reduce operational efficiency and overall agricultural output.
- **High Capital Barrier for Advanced Machinery**
  Premium autonomous equipment remains financially unviable for FPOs and cooperative clusters.
- **Limited Access to Smart Farming Infrastructure**
  Absence of scalable, affordable robotic systems prevents rural regions from transitioning to precision agriculture.

Gap in Existing Solutions

- High-end automation providers focus on large-scale farms, not cooperative-based rural deployment.
- Continued dependency on manual labor due to lack of decentralized mechanization models.
- Existing mechanization schemes lack modular, shared robotic infrastructure solutions.

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03.
Our Solution

Our Solution

Cluster-Based Robotic Farming Infrastructure for Government & Cooperatives

Farm2Future enables State Governments, District Agriculture Departments, and FPOs to deploy shared robotic farming systems across rural clusters — increasing mechanization without requiring individual farmers to invest in expensive machinery.

What We Deliver

- **Decentralized Robotic Units**

  Modular, task-specific agricultural robots deployed at FPO, PACS, or cooperative level for shared usage.

- **Shared Mechanization Model**

  A centralized booking and allocation system allowing multiple farmers within a cluster to access robotic services efficiently.

- **Multi-Task Automation**

  Robots designed for planting, precision spraying, weeding, soil monitoring, and crop inspection.

- **Operational Data & Monitoring**

  Field-level performance tracking to measure productivity improvements, acreage coverage, and yield impact.

- **Deployment, Training & Maintenance Support**

  End-to-end implementation including operator training, servicing, and performance optimization.

Our Solution

Target Audience:

Farm2Future partners with government bodies and rural institutions to deploy shared robotic farming infrastructure at scale.Primary Customers (Paying Entities):

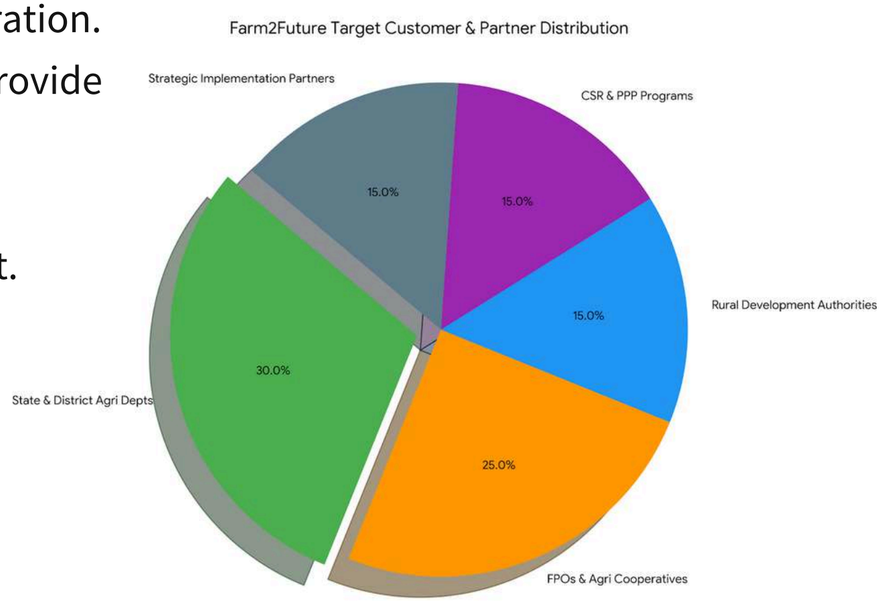
- State and District Agriculture Departments seeking to increase farm mechanization penetration.
- Government-backed Farmer Producer Organizations (FPOs) and Cooperatives aiming to provide shared robotic infrastructure.
- Rural Development Boards and Smart Village initiatives promoting agri-automation.
- CSR programs and Public-Private Partnerships focused on rural productivity enhancement.

End Beneficiaries (Impact Layer):

- Small and marginal farmers operating under FPOs and cooperative clusters.
- Rural farming communities with low mechanization access.
- Regions transitioning from traditional to technology-enabled agriculture.

Strategic Partners:

- Agricultural universities and Krishi Vigyan Kendras (KVKs).
- Agri-skilling and rural training institutions.
- Panchayat-level digital transformation initiatives.
- Agri-service entrepreneurs operating under government-supported schemes.



Our Fleet of AgriBots Built for the Field.

Agribot:

A multi-functional rental robot for planting, weeding, and crop monitoring.

AgriDrone:

For precision spraying and aerial crop monitoring.

AgriArm:

For greenhouse automation (pruning, sorting, planting).

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04.
The Architecture

Agribot— The Core Farming Assistant:

Overview:

Agribot is Farm2Future's flagship autonomous robot built to handle essential field operations such as planting, weeding, soil monitoring, and crop maintenance. It enables farmers to automate repetitive and labor-intensive tasks while keeping costs low through a flexible rental model.

Key Features:

- **Multi-Functional Design:** Handles ploughing, seeding, and watering operations at one go efficiently.
- **Manual & Assisted Modes:** Farmers can switch between remote control and guided navigation.
- **Smart Obstacle Detection:** Equipped with ultrasonic and infrared sensors to avoid field obstacles.
- **Rugged Design:** Built for diverse terrains with durable chassis and water-resistant casing.
- **Battery Efficient:** Runs on a rechargeable lithium-ion system ensuring 6–8 hours of field work.

Impact:

Agribot replaces traditional manual labor with automated precision farming, reducing costs, improving yields, and addressing the persistent labor shortage problem faced by small and medium-scale farmers.

AgriDrone — The Eye in the Sky:

Overview:

AgriDrone is an aerial farming assistant that combines drone technology with AI-based analytics to monitor crop health, detect pest infestations, and conduct precision pesticide spraying.

Key Features:

- **Smart Imaging:** Captures NDVI and multispectral data to analyze plant health.
- **Precision Spraying:** Automated nozzles ensure targeted and uniform pesticide use.
- **AI Analytics Dashboard:** Converts aerial imagery into actionable insights for farmers.
- **Lightweight & Portable:** Easy to assemble, operate, and maintain.
- **Rental Service:** Available for short-term use during critical crop growth stages.

Impact:

AgriDrone minimizes pesticide wastage, enhances field visibility, and empowers farmers to make data-driven decisions that improve productivity and sustainability.

AgriArm — The Greenhouse Automator

Overview:

AgriArm is a robotic manipulator system tailored for controlled environments like greenhouses, nurseries, and indoor farms. It automates labor-heavy tasks such as pruning, sorting, watering, and planting.

Key Features:

- **Robotic Arm Precision:** Equipped with multi-axis motion for delicate handling of plants.
- **Camera Vision System:** Identifies crops and performs actions based on growth stage.
- **Programmable Tasks:** Can be customized for various greenhouse operations.
- **Energy Efficient:** Consumes minimal power while providing consistent performance.
- **Sustainable Design:** Uses recycled and modular materials for easy maintenance.

Impact:

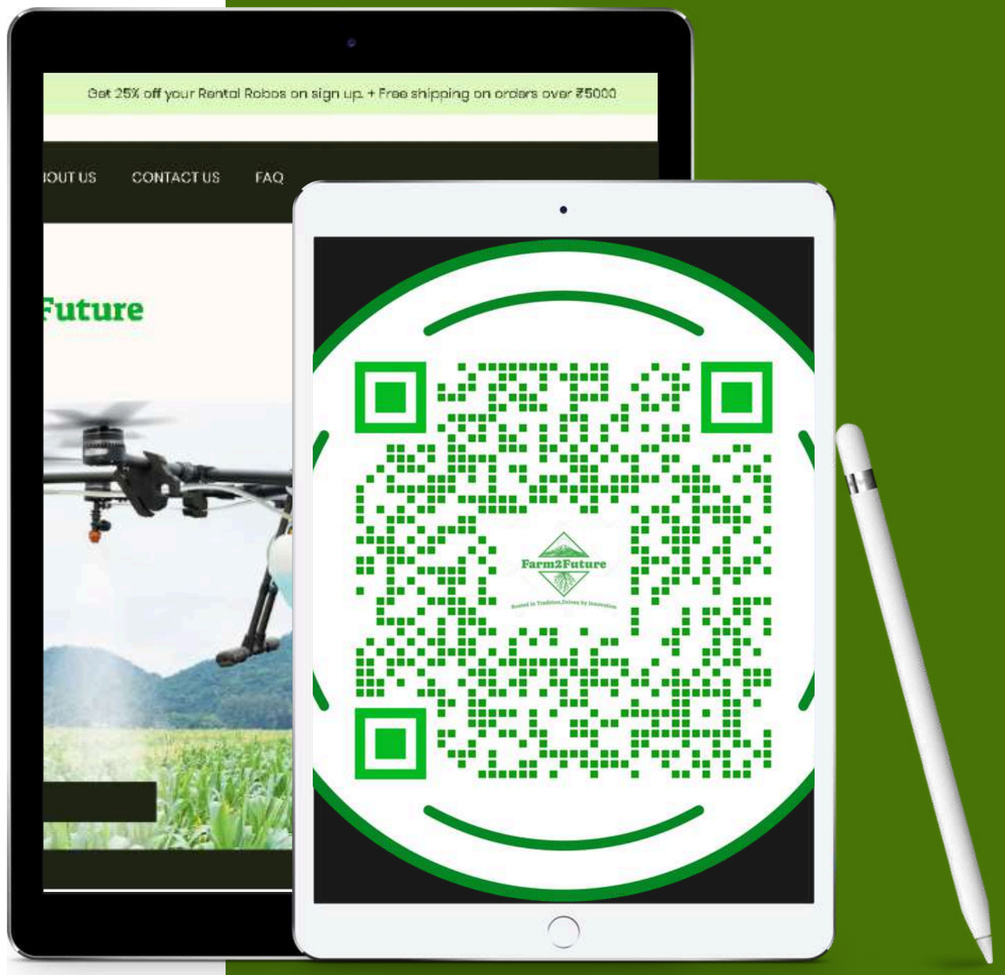
AgriArm ensures consistent quality in greenhouse operations, reduces manual dependency, and increases productivity by up to 35%, making it a vital component in sustainable indoor farming.

Visit Our Website

The **Farm2Future** website serves as the central platform connecting farmers with our products, services, and innovations in modern agriculture. It is designed to be user-friendly and multilingual, ensuring accessibility for farmers from diverse regions.

Key Features:

- **Multilingual Support:** Available in English, Tamil, and Hindi, allowing farmers across India to navigate the platform easily.
- **Agricola – The AI Chatbot:** An intelligent assistant that provides crop recommendations, farming tips, and guidance on selecting the right robotic solutions.
- **Services Section:** Farmers can rent or purchase robots such as Agribot, AgriDrone, and AgriArm directly through the website.
- **Resource Hub:** Offers tutorials, product demonstrations, and articles that promote sustainable and technology-driven farming.
- **Community Blog Page:** A dedicated space where farmers can share their traditional farming practices, innovations, and experiences. High-quality or popular posts can be monetized, helping farmers earn from their knowledge while preserving valuable agricultural wisdom.



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Technology Stack

Technology Stack

AgriBot (Ground Robotics Platform)

- ESP32 Microcontroller (WiFi-enabled control & automation)
- High-Torque DC Motors for field mobility
- Motor Driver Modules for precision control
- Pump Motor System for spraying operations
- Custom Seeding Mechanism for controlled seed dispensing
- Rechargeable Battery Pack with power management
- IoT-ready architecture for future remote monitoring

AgriDrone (Aerial Precision Platform)

- Flight Controller (e.g., Pixhawk / APM-based system)
- Brushless Motors with ESC
- Li-Po High-Capacity Battery
- GPS Module for autonomous navigation
- Onboard Spraying Tank & Nozzle System
- Remote Controller + Telemetry Module
- IMU & Stabilization Sensors

AgriArm (Robotic Manipulation System)

- Arduino Microcontroller
- 6-DOF Servo Motor Mechanism
- Metal Chassis Structure (Industrial-grade frame)
- PWM-based precision motion control
- End-effector gripping mechanism
- Modular design for harvesting & object handling

Digital Platform / Website

- Hosted Web Platform (Wix-based deployment)
- Cloud-based documentation access
- Zapier - Custom Chatbot
- IoT integration ready (Future dashboard support)
- Online demo & technical documentation hosting

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05.
Industry Overview

Industry Overview

Current Industry Scenario:

The agricultural robotics sector is expanding rapidly due to labor shortages, sustainability needs, and smart farming initiatives. Globally, it was valued at USD 10.4 billion in 2024 and is expected to reach USD 38 billion by 2033 (CAGR 15.4%). In India, the market stood at USD 630 million in 2023 and is projected to hit USD 2.65 billion by 2030, growing at a strong 22.8% CAGR.

Business Space:

Farm2Future operates in the agri-tech infrastructure segment, enabling government bodies, FPOs, and cooperatives to deploy scalable robotic farming systems across rural clusters.By combining modular robotics, shared mechanization models, and field-level data insights, Farm2Future supports large-scale agricultural modernization and productivity enhancement.

Industry Size:

- Global Agricultural Robotics Market: $26 billion by 2028, CAGR 19%.
- Indian AgriTech Market: ₹1.2 lakh crore by 2030, driven by automation and data-based solutions.

Industry Overview

Revenue Model

Revenue Model

Farm2Future operates through pilot-based institutional partnerships with government bodies, FPOs, and cooperatives.

1. Pilot Deployment Contracts

- Small-scale robotic deployment at FPO / cluster level
- Initial pilot agreement (6–12 months)
- Includes installation, operator training, and onboarding

Revenue Nature: Upfront capital revenue

Paid per cluster / per FPO

2. Annual Service & Maintenance Agreements

- Recurring service contracts post pilot
- Maintenance, software updates, and technical support
- Performance monitoring and reporting

Revenue Nature: Recurring predictable income

1–3 year renewable agreements

3. Pay-Per-Use (Cluster Level)

- Pay-per-acre or per-task usage model
- Revenue collected through FPOs or cooperatives
- Shared robotic access within farmer groups

Revenue Nature: Variable operational income

Scales with acreage coverage

Currently, Farm2Future does not have any established revenue figures, nor are there customer metrics available. The startup is still in its early phases, focusing on building market presence and implementing marketing strategies to acquire customers. As such, there are no specific revenue or customer numbers to report at this stage and LTV, Revenues are given with theoretical analysis in the next slide.

Industry Overview

Revenue Model

Operational Assumptions (Per Robot)

- Average operational days per month: 20 days
- Average coverage per day: 3 acres
- Total acreage covered per month: 60 acres
- Average charge per acre (via FPO): ₹900

Monthly Revenue per Robot =
60 acres × ₹900 = ₹54,000
Annual Revenue per Robot = ₹6,58,000

Cluster-Level Economics

Assuming:

- 3 Robots per Cluster
- 1 FPO managing operations

Monthly Cluster Revenue:
₹54,000 × 3 = ₹1,62,000
Annual Cluster Revenue:
₹1,62,000 × 12 = ₹19,44,000

Cost Assumptions (Startup Stage)

- Approximate build cost per robot: ₹3,00,000, So 3 robots = ₹9,00,000 capital per cluster
- Maintenance & operational cost per robot per year: ₹1,50,000, So 3 robots = ₹4,50,000/year

Cluster Annual Revenue: ₹19,44,000
Cluster Annual Operating Cost: ₹4,50,000
Cluster Annual Contribution:
₹19,44,000 − ₹4,50,000 = ₹14,94,000

Scaling Example

If deployed in 3 clusters:
Annual Revenue = ₹14,94,000 × 3 = ₹44,82,000
If deployed in 5 clusters:
Annual Revenue = ₹74,70,000

1 Cluster = 25–50 Farmers

- **Approx 100–150 acres combined**
- **3 Robot per cluster (initial pilot)**

Long-Term Value

- Recurring usage revenue
- Annual maintenance contracts
- Expansion through additional cluster deployment
- High scalability with controlled capital expenditure

Industry Overview

Competition

- **Market Position:** Farm2Future exists at the intersection of agriculture and technology, targeting small and medium-scale farmers who lack access to high-end, costly robotic solutions.
- **Key Advantages of Farm2Future:**

  1. Affordable Cluster-Based Access

  Eliminates high ownership costs by deploying robotic systems at the FPO/cooperative level, enabling shared access for small and medium farmers.

  2. Scalable Government-Ready Model

  Designed for village, district, and state-level deployment, easily integrating with agricultural mechanization schemes and public-private partnerships.

  3. End-to-End Service & Local Support

  Provides deployment, training, maintenance, and operational monitoring to ensure long-term usability and reliability in rural environments.

  4. Data-Driven Productivity & Impact Tracking

  Generates field-level usage and performance data, helping government bodies measure productivity improvements and ensure transparent utilization of funds.
- **Competitors:**
  - **Established Players:** John Deere, Mahindra, and TAFE dominates the market with high-cost ownership models and advanced farming equipment.
  - **Emerging Agri-Tech Startups:** Players such as Small Robot Company, Naïo Technologies, and AgXeed offer innovative solutions but may focus on niche or regional markets.
  - **Traditional Equipment Providers:** Brands like Mahindra and TAFE that provide conventional equipment at competitive pricing but lack advanced robotics.

Industry Overview

Competition

Differentiation:

- Farm2Future's cluster-based deployment model eliminates high ownership costs, enabling FPOs and cooperatives to access advanced robotic infrastructure without heavy capital investment.
- Focused on empowering small and medium-scale farmers through shared access, addressing segments often underserved by large mechanization companies.
- A strong emphasis on affordability, sustainability, and productivity enhancement tailored specifically to Indian agricultural conditions and regional crop needs.
- Aligned with Government Mechanization & Rural Development Programs:
  Designed to integrate with state and central agricultural schemes, enabling structured deployment through FPOs, cooperatives, and district-level initiatives.
- Data-Driven Monitoring & Impact Reporting:
  Provides operational insights, usage analytics, and productivity data that help government bodies track outcomes, improve policy decisions, and measure return on public investment.

Our Edge: Farm2Future offers flexibility, affordability, and accessibility—a combination unmatched in the current market landscape.

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06.
Features/USP

Features/USP

Core Features of Farm2Future

1. **Integrated Multi-Robot Ecosystem**

Farm2Future combines three complementary robotic systems:

- **AgriBot** – Autonomous ground operations (seeding, spraying, mobility tasks)
- **AgriArm** – Precision handling and mechanized support functions
- **AgriDrone** – Aerial monitoring, spraying, and crop surveillance

Unlike single-function competitors, this integrated ecosystem enables support across multiple stages of the farming cycle, increasing utilization and improving overall productivity within each cluster.

2. **Cluster-Based Deployment Architecture**

Instead of targeting individual ownership, Farm2Future deploys robotic infrastructure at the FPO/cooperative level, serving 25–50 farmers per cluster.

This approach:

- Reduces per-farmer cost
- Increases robot utilization rates
- Ensures predictable revenue flow
- Makes adoption viable in small landholding regions

The model transforms robotics from a luxury product into shared rural infrastructure.

3. **Usage-Based Revenue Model (₹800 per Acre)**

Transparent **pay-per-acre** pricing eliminates financial barriers and aligns revenue with actual productivity.

For institutions and government bodies, this ensures:

- Measurable ROI
- Clear cost justification
- Scalable budgeting

Revenue is directly linked to operational deployment rather than one-time equipment sales.

4. **End-to-End Service & Maintenance**

Farm2Future handles:

- Deployment
- Scheduling across farmers
- Operator training
- Preventive maintenance
- Real-time monitoring

Farmers and FPOs do not need technical expertise. The company ensures smooth and continuous operations.

5. **Data-Driven Impact Monitoring**

Each deployment generates operational data:

- Acres covered
- Input optimization
- Usage frequency
- Productivity trends

This enables measurable impact reporting for government programs and institutional stakeholders.

Features/USP

Unique Selling Proposition

Operating Model Structure

Farm2Future operates a cluster-based robotic infrastructure model deployed through FPOs and cooperatives.

Each cluster **(25–50 farmers)** is supported by **3 robots**, enabling high utilization per asset and recurring acreage-based revenue rather than one-time sales.

Revenue Model: **₹900 per acre.**

Competitor Model Comparison

Large OEMs (Mahindra, TAFE, John Deere):

Ownership-driven sales (₹5–30 lakh per unit). Revenue dependent on capital purchases. Limited recurring operational engagement.

Agri-Robotics Startups:

Product-focused or pilot-based deployments. Higher R&D-driven pricing. Limited rural service infrastructure.

Local Contractors:

Provide tractor/spraying rentals. Task-specific, non-automated, no robotics integration or performance data.

Utilization & Capital Efficiency Advantage

Ownership Model:

1 machine → 1 farmer → limited seasonal usage.

Farm2Future Model:

1 robot fleet → 25–50 farmers → multiple acre-based transactions per season.

Higher asset rotation → Higher revenue per unit → Better capital efficiency.

Features/USP

SWOT Analysis

Strengths

- Affordable rental model removes capital barrier for smallholders.
- Localized support hubs and quick service response.
- Integrated AI advisory (Agricola) and multilingual website support.
- Strong technical founding team with proven prototypes and award recognition.

Weaknesses

- Early-stage brand with limited fleet and operational scale.
- Dependence on regional logistics; higher initial OPEX for hubs.
- Need for working capital to expand fleet and build spare inventory.

Features/USP

SWOT Analysis

Opportunities

- High growth in India's agri-robotics market (strong CAGR) and unmet demand among small/medium farms.
- Government schemes and subsidies for farm mechanization.
- Monetizing farmer content, training, and data-driven advisory as additional revenue streams.
- Partnerships with FPOs, local OEMs, and agri-input companies.

Threats

- Entry by large OEMs into rental/managed-service models.
- Rapid technology evolution requiring ongoing R&D.
- Macroeconomic shocks (poor monsoon, crop price falls) that lower farmer spend.
- Supply-chain disruptions and component cost inflation.

The Team Behind Us

Teamwork Is The Key

Sanjith Sharmaa R
Co-Founder & CTO

Expertise: Bot building, Tech Enthusiast
Success: Skilled in creating bots and engaging with potential supporters, contributing to the project's growth and development.

Adithya K
Co-Founder & CEO

Expertise: Bot building, coding, and website development
Success: Leading the team and driving the technical side of the business, with a keen interest in robotics and coding

Hariharan R S
Co-Founder & R&D Head

Expertise: Coding and 3D modeling
Success: Dedicated to enhancing the coding and design of robotic systems, ensuring efficiency and innovation.

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Appendices

Appendices

Milestones

2023 Milestones:

- Agribot Inception: As a team, we developed the idea for Agribot to address challenges in small and medium-scale farming.
- **National-Level Selection:** Our Agribot idea was selected in the first round of a national-level competition, earning a **₹10,000 cash award** through the **INSPIRE Award 2022-23.**

2024 Milestones:

- **November 8, 2024:**
  - Selected as a finalist in **Puthulir 2024** District level selection, hosted by the Centre for Entrepreneurship Development, Anna University, Chennai.
- **November 14, 2024:**
  - Victory at **Puthulir 2024** Finals:
    - Board-Wise Award Winner - CBSE Category.
    - Honored with a ₹**15,000 cash prize**, by the Hon'ble Education Minister of Government of Tamil Nadu, **Thiru Anbil Mahesh Poyyamozhi.**
    - Recognized for its innovative approach to leveraging rental Agribots for sustainable and efficient farming practices.

Appendices

Milestones

2025 Milestone:

- January 9, 2025: 2nd Runner-Up under the project theme Responsible Consumption and Production at **SAI TECH S2T EDITION 5.0, held at SAIRAM INSTITUTIONS CAMPUS.**
- Award: **₹5,000 cash prize** in the Senior Scientist category.
- **November 5, 2025:** Winner under school category at Nation-wide Ideathon 5.0 held at **Kalinga University Raipur**. Award: **₹25,000 cash prize** + National Level Certificate

Appendices

References/Links

- FAO Precision Agriculture Guidelines
- ICAR (Indian Council of Agricultural Research)Publications
- Agricultural Robotics Research (Springer Link)
- Arduino Documentation

Project Repository & Documentation

All necessary materials, including:

- Source Code
- Circuit Diagrams
- System Architecture
- Hardware Design Files
- Deployment Documentation
- Demo Video

can be found in the official GitHub repository below

→ Farm2Future GitHub

→ Farm2Future Website

THANKS

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